Project: Worked examples
Title: Psychrometrics
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The state of moist air (humidity ratio, relative humidity, dew point, wet-bulb, enthalpy, specific volume, density) from dry-bulb and one other property at a stated pressure or altitude, and one air-handling process from that state with its heat and moisture balance: sensible heating or cooling, cooling with dehumidification, steam humidification, adiabatic humidification, or mixing with a second stream. p_{ws} from Hyland–Wexler (over water ≥ 0 °C, over ice below); $W = 0.621945\, p_w/(p - p_w)$; $\varphi = p_w/p_{ws}$; $h = 1.006\, t + W(2501 + 1.86\, t)$ kJ/kg_da; $v = 0.287042\,(t + 273.15)(1 + 1.607858\, W)/p$; dew point and wet-bulb by bisection (wet-bulb from the adiabatic-saturation relation). Processes on a dry-air mass flow: sensible $Q = \dot{m}\,(h_2 - h_1)$ at constant W; cooling/dehumidification $Q_{total} = \dot{m}(h_1 - h_2) - \dot{m}_w\, h_w$ with condensate $\dot{m}_w = \dot{m}(W_1 - W_2)$; steam $h_2 = h_1 + (W_2 - W_1)\, h_{steam}$; adiabatic humidification at constant h; mixing by mass-weighted W and h.

Ref: ASHRAE Handbook — Fundamentals Ch. 1 — Psychrometrics (ideal-gas relations, processes)
Ref: Hyland & Wexler, ASHRAE Trans. 89(2A), 1983 — Saturation pressure of water and ice
Ref: CIBSE Guide C Section 1 — Properties of humid air (principles)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
altitude	Altitude (used when pressure is blank)	0	m	assumption: Sea-level site.	-500 … 5000 m
t_db	Dry-bulb temperature	28	degC	spec: CIBSE Guide A summer design condition for the site, assumed	-40 … 60 degC
known	Second known property	relative humidity		assumption: RH given.	any
RH	Relative humidity	60		spec: Design condition	0 … 100
m_da	Dry-air mass flow	2.4	kg/s	spec: AHU schedule AHU-01: 2.0 m³/s ≈ 2.4 kg/s dry air	0 … 1000 kg/s
process	Process	cooling with dehumidification		assumption: Chilled-water coil.	any
t_2	Leaving dry-bulb (sensible, cooling)	12	degC	spec: AHU schedule AHU-01 off-coil 12 °C	-40 … 80 degC
RH_2	Leaving relative humidity (cooling, evaporative)	95		assumption: Off-coil 95 % RH (typical contact factor).	0 … 100
h_steam	Steam enthalpy (steam humidification)	2676		default	2500 … 3000

3. Assumptions

A1. Moist air is an ideal mixture of ideal gases; the enhancement factor is taken as 1. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A2. Condensate leaves at the coil leaving dry-bulb; steam adds enthalpy h_steam at negligible mass of dry air. (Process conventions.)
A3. Adiabatic humidification follows a line of constant enthalpy (close to constant wet-bulb). (Evaporative process idealisation.)
A4. altitude (Altitude (used when pressure is blank)) is taken as 0 m. (Sea-level site.)
A5. known (Second known property) is taken as relative humidity. (RH given.)
A6. process (Process) is taken as cooling with dehumidification. (Chilled-water coil.)
A7. RH_2 (Leaving relative humidity (cooling, evaporative)) is taken as 95. (Off-coil 95 % RH (typical contact factor).)

4. Working

```
[2] p = 101.325 (1 - 2.25577e-5 z)^5.2559                    ASHRAE Ch. 1 eq. 3
= z = 0 m
= 101.32 kPa
```
Barometric pressure (standard atmosphere)
```
[6] p_ws_1 = p_ws(t) Hyland-Wexler (entering)
= t = 28 °C
= 3.7822 kPa
```
Saturation pressure, state 1
```
[7] p_w_1 = p W/(0.621945 + W)
= W = 0.014249
= 2.2693 kPa
```
Vapour pressure, state 1
```
[8] t_1_out = dry-bulb (entering)
=
= 28 degC
```
Dry-bulb, state 1
```
[9] W_1 = 0.621945 p_w/(p - p_w)
= 2.2693 / (101.32 - 2.2693)
= 14.249
```
Humidity ratio, state 1 (g/kg dry air)
```
[10] RH_1 = p_w / p_ws
= 2.2693 / 3.7822
= 60
```
Relative humidity, state 1 (%)
```
[11] phi_1 = p_w / p_ws
= 2.2693 / 3.7822
= 60 %
```
Relative humidity, state 1, as a percentage quantity (the same value as RH_1; a plain-number output cannot be linked to a percentage input)
```
[12] t_dp_1 = p_ws(t_dp) = p_w, bisection
= p_w = 2.2693 kPa
= 19.514 degC
```
Dew point, state 1
```
[13] t_wb_1 = adiabatic-saturation relation, bisection
= t = 28, W = 0.014249
= 22.074 degC
```
Wet-bulb, state 1
```
[14] h_1 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 28 + 0.014249 × (2501 + 1.86 × 28)
= 64.546
```
Enthalpy, state 1 (kJ/kg dry air)

[15] v_1 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 101.32 kPa
= 0.87267 m³/kg
Specific volume, state 1 (per kg dry air)
[16] rho_1 = (1 + W)/v
=
= 1.1622 kg/m³
Moist-air density, state 1
[21] p_ws_2 = p_ws(t) Hyland–Wexler (leaving)
= t = 12 °C
= 1.4026 kPa
Saturation pressure, state 2
[22] p_w_2 = p W/(0.621945 + W)
= W = 0.0082878
= 1.3325 kPa
Vapour pressure, state 2
[23] t_2_out = dry-bulb (leaving)
=
= 12 degC
Dry-bulb, state 2
[24] W_2 = 0.621945 p_w/(p − p_w)
= 1.3325 / (101.32 − 1.3325)
= 8.2878
Humidity ratio, state 2 (g/kg dry air)
[25] RH_2 = p_w / p_ws
= 1.3325 / 1.4026
= 95
Relative humidity, state 2 (%)
[26] phi_2 = p_w / p_ws
= 1.3325 / 1.4026
= 95 %
Relative humidity, state 2, as a percentage quantity (the same value as RH_2; a plain-number output cannot be linked to a percentage input)
[27] t_dp_2 = p_ws(t_dp) = p_w, bisection
= p_w = 1.3325 kPa
= 11.225 degC
Dew point, state 2
[28] t_wb_2 = adiabatic-saturation relation, bisection
= t = 12, W = 0.0082878
= 11.553 degC
Wet-bulb, state 2
[29] h_2 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 12 + 0.0082878 × (2501 + 1.86 × 12)
= 32.985
Enthalpy, state 2 (kJ/kg dry air)
[30] v_2 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 101.32 kPa
= 0.81856 m³/kg
Specific volume, state 2 (per kg dry air)
[31] rho_2 = (1 + W)/v
=
= 1.2318 kg/m³
Moist-air density, state 2

[32] m_w = □ (W₁ - W₂)
= 2.4 × (14.249 - 8.2878) g/kg
= 0.014306 kg/s
Condensate
[33] h_w = 4.186 t₂ (condensate enthalpy)
= 4.186 × 12
= 50.232
Condensate enthalpy, kJ/kg
[34] Q_total = □ (h₁ - h₂) - □_w h_w
= 2.4 × (64.546 - 32.985) - 0.014306 × 50.232
= 75.027 kW
Total cooling load
[35] Q_sensible = □ (h(t₂, W₁) - h₂)
= 2.4 × (48.026 - 32.985)
= 36.098 kW
Sensible part (temperature change at W_1 ... approximated at W_1)
[36] Q_latent = □ (h₁ - h(t₂, W₁))
= 2.4 × (64.546 - 48.026)
= 39.648 kW
Latent part
[37] SHR = Q_sensible / (Q_sensible + Q_latent)
=
= 0.47656
Sensible heat ratio

5. Results

SYMBOL	VALUE	DESCRIPTION
t_1_out	28 degC	Dry-bulb 1
p_ws_1	3.7822 kPa	p_ws 1
p_w_1	2.2693 kPa	p_w 1
W_1	14.249	W 1 (g/kg)
RH_1	60	RH 1 (%)
phi_1	60 %	φ 1 (RH as a % quantity, for linking)
t_dp_1	19.514 degC	Dew point 1
t_wb_1	22.074 degC	Wet-bulb 1
h_1	64.546	h 1 (kJ/kg)
v_1	0.87267 m³/kg	v 1
rho_1	1.1622 kg/m³	ρ 1
t_2_out	12 degC	Dry-bulb 2
p_ws_2	1.4026 kPa	p_ws 2
p_w_2	1.3325 kPa	p_w 2
W_2	8.2878	W 2 (g/kg)
RH_2	95	RH 2 (%)
phi_2	95 %	φ 2 (RH as a % quantity, for linking)
t_dp_2	11.225 degC	Dew point 2

SYMBOL	VALUE	DESCRIPTION
t_wb_2	11.553 degC	Wet-bulb 2
h_2	32.985	h 2 (kJ/kg)
v_2	0.81856 m³/kg	v 2
rho_2	1.2318 kg/m³	ρ 2
m_w	0.014306 kg/s	Condensate
Q_total	75.027 kW	Total cooling
Q_sensible	36.098 kW	Sensible cooling
Q_latent	39.648 kW	Latent cooling
SHR	0.47656	Sensible heat ratio

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

WARNING — input.h_steam: h_steam (Steam enthalpy (steam humidification)) was not entered; the template default 2676 is used.

8. Verification

Template bs.psychrometrics@1.1.0 Content hash 019c21a8f58b1d42 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0